CSE 598 008
EECS 498 003
Lab 7

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Based on slides by Ivan Wei (Fall 2025).

Reminder

- Problem Set 3 (Chapter 5) due October 26, 11:59pm
- Start looking for Project 1 partners (released after PS3)
- No lecture on Tuesday, October 20 (study break)
- Midterm will be on Thursday, October 22, 6pm – 8pm, FMCRB 1050
  - No lecture that day
  - Closed book; one double-sided cheat sheet allowed, 10pt minimum
  - Covers everything up to Chapter 4 (excluding distributed systems)
  - Review session on practice midterms: Tuesday, October 20, time and place TBD
- Midterm evaluations are due on October 22, please fill them in

Today

- Midterm review
  - A few true or false questions on Chapters 1–4
  - Feel ready for the midterm? Work on the distributed system exercise instead
- Extra office hours for the rest of the lab
  - Bring any questions: PS2 grading, PS3, or anything from Chapters 1–4

Distributed System Exercise

- Our midterm and PS3 overlap, so this exercise is optional today
- Feel ready for the midterm? Work on it while we review
  - It helps with PS3 and with later class content
- Distributed Lock with Network
  - Starter code: `lab07.dfy` on the course GitHub, with `UtilitiesLibrary.dfy`
- We will continue it in Lab 8 and go over the answer

True or False

- The following verifies:

```
lemma ContrivedLemma(x: nat, y: nat)
    requires x <= y
    ensures
        && y == x + 1 <==> y - x == 1
        && y <= x <==> x == y
{}
```

True or False

- Consider a crawler, starting at (3, 0), that moves north every second
- We can write a state machine, as typical of this course, that models this crawler

True or False

- Consider a crawler, starting at (3, 0), that can move north. There is a dangerous hole at the origin with radius 2
- Suppose Dafny forgets that the crawler starts at (3, 0)
  - Dafny can prove: if the crawler is not in the hole, then after a move, it is not in the hole
  - Dafny can prove: if the crawler is at x-coordinate 3, then after a move, it is at x-coordinate 3

True or False

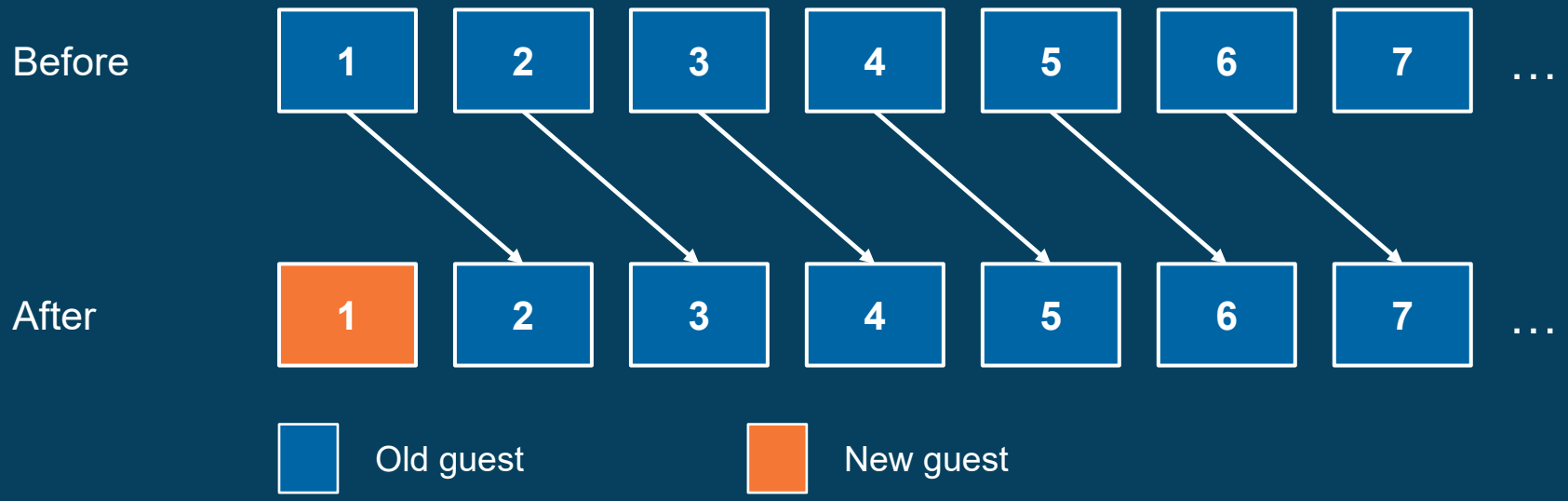
- Ghost functions and predicates can be used in non-ghost functions and predicates
- Non-ghost functions and predicates can be used in ghost functions and predicates

Extra - Hilbert's Hotel

- Hilbert's Hotel has infinitely many rooms, numbered 1, 2, 3, …
- Tonight, every room is taken
- One more guest arrives. Can the hotel give them a room, without sending anyone away?

Extra - Hilbert's Hotel

- Yes! Every guest moves from room n to room n + 1
- Room 1 is now free for the new guest

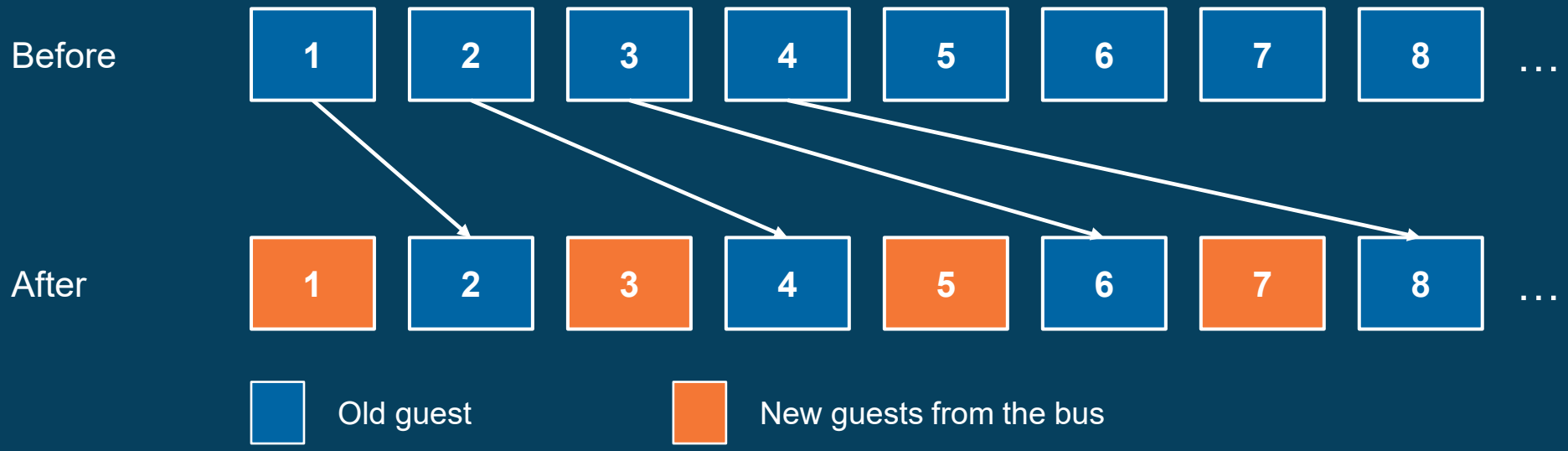


Extra - Hilbert's Hotel

- The hotel is full again
- A bus arrives with infinitely many new guests, numbered 1, 2, 3, …
- Can the hotel give every one of them a room?

Extra - Hilbert's Hotel

- Yes! Every guest moves from room n to room 2n
- All the odd rooms are now free: new guest k goes to room 2k − 1



https://en.wikipedia.org/wiki/Hilbert's_paradox_of_the_Grand_Hotel